MAS · Integration Techniques and Applications — Patterns and Solution Strategies

Source: marking keys for 44 past exam questions from 2016–2025 + exam reports. Original questions in `topics/05_Integration.pdf`. Numerical examples checked with sympy. Review status: one round of independent review completed and revisions made according to feedback (see `Review_log.md`).

Marker reminders

- When the antiderivative involves ln, include absolute value signs (2022 report). In recent keys this is counted in the "antiderivative" mark; if the absolute value signs are missing, the full mark may not be awarded; when the argument is always positive (e.g. $\ln(x^2 + 2)$), the key notes they may be omitted.
- Indefinite integrals require + c (almost every partial fractions question has this 1 mark).
- Basic algebra skills: over 30% of candidates expanded $(\sqrt{3}\tan u + 1)^2$ incorrectly (2018 report); incorrectly applied the distributive law, writing $\sqrt{(2 - \sin^2\theta)}$ as $\sqrt{2} - \sin\theta$ (2021 report, F-Q8(c)); wrote the numerator expression incorrectly after finding a common denominator in partial fractions (2024 report).
- When integrating using a given substitution, after substitution simplify fully before integrating (2020, 2023 reports).
- Volume formulas must include π; all integral expressions must include dx/dy and limits (the keys for 2022A-Q17 and 2021A-Q13 combine "limits + correct notation" into 1 mark).

Pattern 1: Definite integrals with a given substitution ★ CF almost every year (only 2022 not examined in 2016–2025), 4–7 marks

Fixed five steps (the 5 marks in the key are basically these five steps)

1. Find dx from the substitution: $u = 1 - x \Rightarrow dx = -du$; $x = 2\sin\theta \Rightarrow dx = 2\cos\theta\, d\theta$.
2. Change the limits: $x = 0 \to u = 1$, $x = 1 \to u = 0$.
3. Write the integrand completely in terms of the new variable and simplify (commonly use $1 - \sin^2\theta = \cos^2\theta$, $1 + \tan^2 u = \sec^2 u$).
4. Find the antiderivative.
5. Substitute the limits and give the exact value.

Three types of substitution

- Linear or radical: $u = 1 - x$, $u = \sqrt{(x + 2)}$, $x = 119u + 1$ (2020F-Q7, 2021F-Q3, 2023F-Q3, 2024F-Q3).
- Power functions via reverse chain rule: $u = \sin 2x$, $u = \cos 2x$ (2016F-Q5(a), 2018F-Q5).
- Trigonometric substitution: $x = a\sin\theta$ to handle $\sqrt{(a^2 - x^2)}$; $x = a\tan u$ to handle $(x^2 + a^2)$.

Trigonometric substitution example (2019F-Q6): $\int_0^{\sqrt{3}} \sqrt{(1 - x^2/4)}\, dx$, let $x = 2\sin\theta$, limits become $0 \to \pi/3$ $\Rightarrow \int_0^{\pi/3} 2\cos^2\theta\, d\theta = \int(1 + \cos 2\theta)d\theta = \pi/3 + \sqrt{3}/4$.

Past questions: 2016F-Q5, 2017F-Q3, 2018F-Q5, 2018F-Q9, 2019F-Q6, 2020F-Q7, 2021F-Q3 (choose the substitution yourself), 2021F-Q8(c), 2023F-Q3, 2023F-Q7(b), 2024F-Q3, 2025F-Q6. Indefinite integral version (2016A-Q9, $\int x(1 + x)^n\, dx$): there is no change-of-limits step; at the end substitute back to x.

Pattern 2: Integration after simplifying with trigonometric identities (CF 3–5 marks)

Common identities

- $\cos^2 x = \frac{1}{2}(1 + \cos 2x)$, $\sin^2 x = \frac{1}{2}(1 - \cos 2x)$;
- $(\sin x + \cos x)^2 = 1 + \sin 2x \Rightarrow \int_0^{\pi/2} = \pi/2 + 1$ (2022F-Q3);
- product-to-sum: $2\sin A\cos B = \sin(A + B) + \sin(A - B) \Rightarrow \int_0^{\pi/2} 6\sin(5x/2)\cos(x/2)\, dx = 4$ (2019F-

Q1);

- $\tan^2 x = \sec^2 x - 1$ (2016F-Q5(b));
- $\int_0^{\pi} (4\cos^2 x - \sin x)\,dx = \int (2 + 2\cos 2x - \sin x)dx = 2\pi - 2$ (2020F-Q1).

Past questions: 2016F-Q5(b), 2019F-Q1, 2020F-Q1, 2022F-Q3.

Pattern 3: Partial fractions ★ CF frequently tested (examined in 8 of 10 years; not in 2017 or 2023)

Standard procedure

1. Set up the form:
    - distinct linear factors: $A/(x - a) + B/(x - b)$;
    - repeated root: $A/(x + 1) + B/(x + 1)^2$ (2024F-Q4); $A/x + B/x^2 + C/(x + 3)$ (2019F-Q3);
    - irreducible quadratic: $A/(x - 2) + (Bx + C)/(x^2 + 2)$ (2021F-Q5, $x^2 + 4$ in 2020F-Q6).
    - When the question specifies the form, follow the given form: 2022F-Q4 requires writing it as $a/(x - 1) + (bx + c)/(x^2 + 3x + 1)$ (this quadratic can actually be factorised; the discriminant is $5 > 0$).
2. After finding a common denominator, equate the numerators and substitute special values or compare coefficients to find the constants ("equivalence of numerators" 1 mark, solving for the constants 1 mark).
3. Integrate term by term: $A/(x - a) \rightarrow A \ln|x - a|$; $B/(x + 1)^2 \rightarrow -B/(x + 1)$; for terms such as $(2x + 3)/(x^2 + 3x + 1)$ where the numerator is exactly the derivative of the denominator $\rightarrow \ln|x^2 + 3x + 1|$; $q/(x^2 + 4) \rightarrow (q/2)\tan^{-1}(x/2)$ (obtained from the derivative of the inverse trigonometric function, 2020F-Q6).
4. Write + c.

Example: $1/(x^2 - 1) = \frac{1}{2}[1/(x - 1) - 1/(x + 1)] \Rightarrow \int = \frac{1}{2} \ln|(x - 1)/(x + 1)| + c$ (2018F-Q6). $(5x + 3)/(x + 1)^2 = 5/(x + 1) - 2/(x + 1)^2 \Rightarrow \int = 5 \ln|x + 1| + 2/(x + 1) + c$ (2024F-Q4). When there is no hint (2025F-Q3): first factorise the denominator yourself (key mark 1).

Past questions: 2016F-Q4, 2018F-Q6, 2019F-Q3, 2020F-Q6(c)(d), 2021F-Q5, 2022F-Q4, 2024F-Q4, 2025F-Q3.

Pattern 4: Area (integration with respect to x or y)

Approach

1. Find intersection points and tangency points as the limits.
2. Choose the variable of integration: when the curve is more conveniently written as $x = g(y)$, integrate with respect to y ($2x = \sin y$ in 2016A-Q13, 2017A-Q12, 2022F-Q7). The key often awards marks for both methods, but integrating with respect to y usually requires only one integral.
3. Area = $\int$(right − left)dy or $\int$(top − bottom)dx.
4. Use symmetry: ellipse 2025A-Q8; 2022A-Q13(c) symmetric about the x-axis, multiply by 2; for the cardioid in 2021F-Q8, the 4 = the 2 from symmetry about the y-axis × the 2 in the upper/lower branch difference $2\sqrt{(2 - x^2)}$.
5. Interpret a definite integral as an area: 2017A-Q16 (after the substitution $u = 2x - a$ it equals the area of some region), 2024A-Q15(c) (triangle area = 2k).
6. Area of a circular segment: solve the circle equation for $y = 4 \pm \sqrt{(4 - (x - 3)^2)}$; the difference between the upper and lower branches = $2\sqrt{(\ldots)}$ (2023F-Q7(a)).
7. Determine whether a curve is a circle: compare the numerical area with πr^2 (2020A-Q16(b)).

Past questions: 2016A-Q13, 2017A-Q12, 2017A-Q16, 2022A-Q13(c), 2024A-Q15(c), 2018A-Q15(b), 2020A-Q16, 2021F-Q8, 2022F-Q7, 2023A-Q18(b), 2023F-Q7, 2024F-Q8(e), 2025A-Q8.

Pattern 5: Volumes of revolution ★ CA frequently tested

Formulas

- About the x-axis: $V = \pi\int_a^b y^2\,dx$; about the y-axis: $V = \pi\int_a^b x^2\,dy$ (first write x^2 as a function of y).
- Region between two curves: $V = \pi\int(y_{outer}^2 - y_{inner}^2)dx$ (not $\pi\int(y_{outer} - y_{inner})^2dx$).
- Torus (doughnut, 2024A-Q11): $x = 3 \pm \sqrt{(4 - y^2)} \Rightarrow V = \pi\int(R^2 - r^2)dy = \int_{-2}^{2} 12\pi\sqrt{(4 - y^2)}\,dy$.

Contextual question types

- Cup volume formula: derive using an integral with letters a, b, h (2017F-Q8).
- Depth when filled to 80%: set $\int_0^h \pi x^2\,dy = 0.8 \times V_{full}$, solve for h (2020A-Q9).
- Related rates of filling: $dV/dt = (dV/dh)\cdot(dh/dt)$ (2025A-Q14(b): find dV/dh from $V = ah^3$).
- Increment formula: $\delta h \approx \delta V \div (dV/dh)$, where $dV/dh = \pi x^2|_{y=h}$ (2023A-Q17(c)).
- The volume integral in 2022A-Q17 is in (a); (b)–(d) are about the increment, flow rate and separation of variables for dV/dt = 300e^{−V/12000} (see the "Rates of Change and Differential Equations" notes).
- Spherical cap (2021A-Q13): from the circle equation get $y^2 = 20x - x^2$, integrate with respect to x from 0 to h.

Marking points: ✓ radius expressed in terms of x or y ✓ correct integral expression (limits + π + square) ✓ calculation (including units, rounding). Past questions: 2016A-Q17(c), 2017F-Q8, 2018A-Q15(c), 2019F-Q8, 2020A-Q9, 2021A-Q13, 2022A-Q17, 2023A-Q17, 2024A-Q11, 2025A-Q14.